GREGORY GOLONKA

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EDUCATION AND SCHOOL HONORS

University of Notre Dame, South Bend, IN — **2021 – 2025**

- GPA: 3.53
- Major: Aerospace Engineering, College of Engineering

Relevant Coursework: Intermediate Thermodynamics, Fluid Mechanics, Differential Equations I & II, Heat Transfer, Theoretical Aerodynamics, Compressible Aerodynamics, Aerospace Structures, Aerospace Dynamics, Gas Turbines and Propulsions, Flight Mechanics and Intro to Design, Mechanisms and Machines, Orbital and Space Mechanics, Senior Design

Montpelier High School, Montpelier, VT — **2017 – 2021**

- GPA: 3.99 (unweighted)
- Member of Valedictorian group, Rensselaer Polytechnic Medal Award, Montpelier Rotary Cohen Merit Scholarship

TECHNICAL SKILLS

- *Security Clearance:* Able to obtain and maintain security clearance
- *Software:* MATLAB, Excel, Python, C++, HTML, Next.js
- *Systems and Tools:* XFLR5, XFoil, SolidWorks, Autodesk Fusion 360, Mathematica, Arduino IDE
- *Engineering Knowledge:* Flight Dynamics, Controls, Flight Stability, Error Propagation, Fluid Dynamics, Aerodynamics, Propulsion Systems
- *Soft Skills:* Technical Documentation, Team Collaboration, Problem Solving, Time Management, Data Analysis

PROJECTS AND RESEARCH

1D Viscous Shock Wave Analysis with Molecular Dissociation, **University of Notre Dame Undergraduate Research** — **2024 – 2025**

- Conducted a study to advance the understanding of shock wave structure and its interaction with reactive systems, with applications in propulsion, detonation physics, and high-speed flow analysis. Focused on analyzing the dissociation of diatomic molecules across a viscous shock structure, with a goal of understanding molecular bond behavior under extreme temperatures in shock waves

RC Plane, **University of Notre Dame Senior Capstone** — **2024 – 2025**

- Designed a radio-controlled airplane to complete a specific mission, focusing on aerodynamic optimization. Utilized XFLR5 and SOLIDWORKS CFD software to analyze and refine the base model, ensuring efficient flight performance. Created a detailed manufacturing plan for construction, addressing material selection, part assembly, and production steps. The design phase emphasized performance predictions and preparation for the next phase of the project, laying the groundwork for successful testing and final adjustments.

Reengining the BGM-109A Tomahawk Cruise Missile, **Gas Turbines and Propulsions Design Project** — **2025**

- Worked with a team to develop a flight performance simulation code to analyze the BGM-109A Tomahawk missile's mission profile and thrust requirements, utilizing the Williams F107-WR-402 engine's on-design and off-design performance data. Optimized a new turbofan engine design to increase the missile's stand-off range, performing an analysis to improve efficiency and performance across varying flight conditions.

Accessibility Robot Design Challenge, **Design Tools II Final Project** — **2024**

- Managed a engineering project to design a prototype RC delivery robot addressing accessibility needs, focusing on payload delivery systems and diverse terrain navigation. Conducted finite element analysis (FEA) in SolidWorks Simulation and motion analysis for component selection.

Experimental Aerodynamics Lab, — **2024**

- Gained hands on experience using the wind tunnel through a series of experiments at Hessert Laboratories on Notre Dame Campus. Examples of these experiments include the calibration of a pressure transducer to convert pressure measurements into digital recordings, and the use of a force balance to calculate the different drag contributions.

WORK EXPERIENCE

Movie Theater Projectionist --**Capitol Theater** — **2023 – 2024**

- Problem solving digital projection equipment to ensure smooth operation during events.

Dorm Supervisor, **Dunne Hall, University of Notre Dame** — **2022 – 2023**

- Work two shifts a week, acting as both front desk help and chef for the in-dorm food service provided to fellow students.

Landscape Technician, **C. Michaud Landscaping** — **2021 – 2024**

- Operated in a coordinated team environment to execute scheduled routes and specialized projects, servicing over 150+ properties weekly.

LEADERSHIP, CAMPUS INVOLVEMENT, AND SERVICE

Vice President – **Dunne Hall, Hall Presidents Council** — **2023 – 2024**

- Managed project planning, funding, and execution of dorm-wide events, facilitating collaboration among community leaders.

Saint Andre Committee Member – **Dunne Hall, University of Notre Dame** — **2022 – 2023**

- Welcomed and supported incoming freshmen, promoting a positive and inclusive community environment.